



Understanding the Resource Usage of High-Energy Physics Applications in HPC Systems

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Motivation

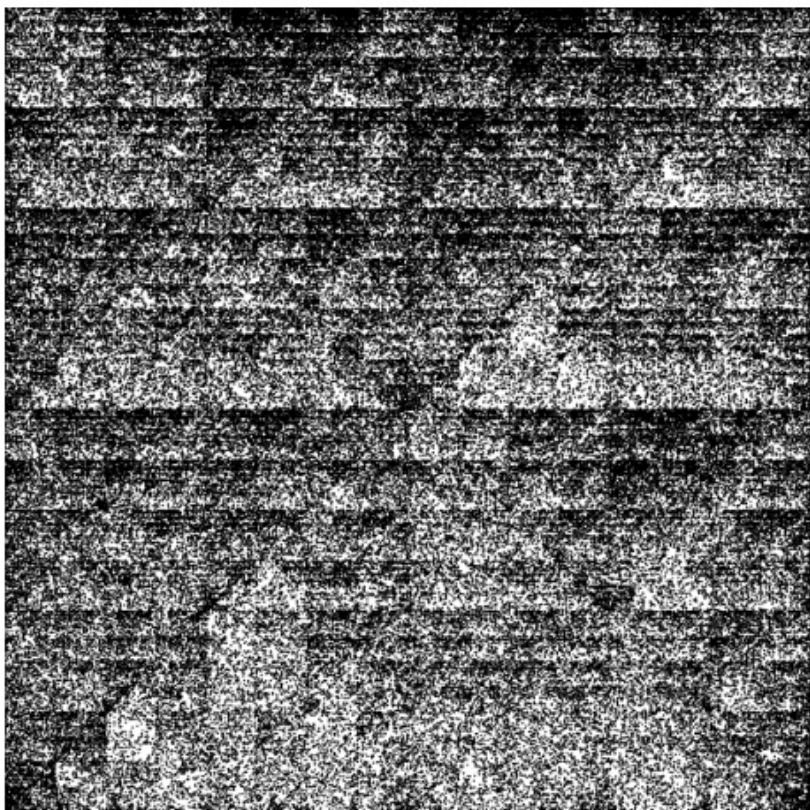
- Scaling scientific applications -> more time, energy, and compute used
- Sustainable computing requires resource usage optimizations
- Resource usage patterns in HPC applications are complex
- Need to profile/fingerprint data to better understand it

Existing Methods

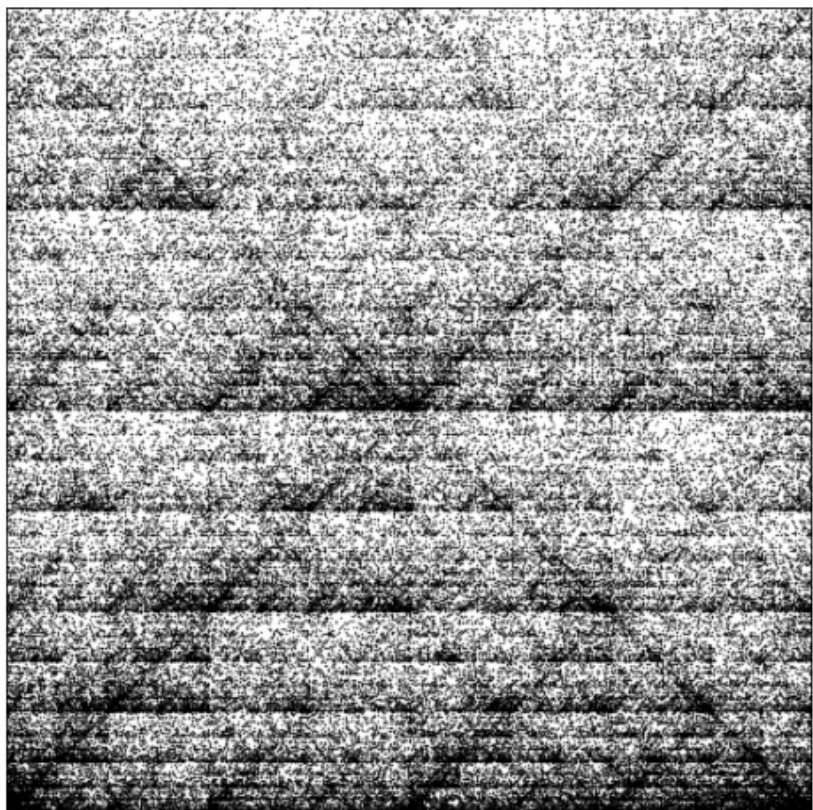
- Time series analysis is prevalent in many scientific domains
 - Weather Data, ECG, EEG, etc.
- Techniques to filter and characterize data
 - Data preprocessing and smoothing
 - Statistical methods (ARIMA, Autocorrelation, etc.)
 - AI/ML Methods

Chaos Game Representation (CGR)

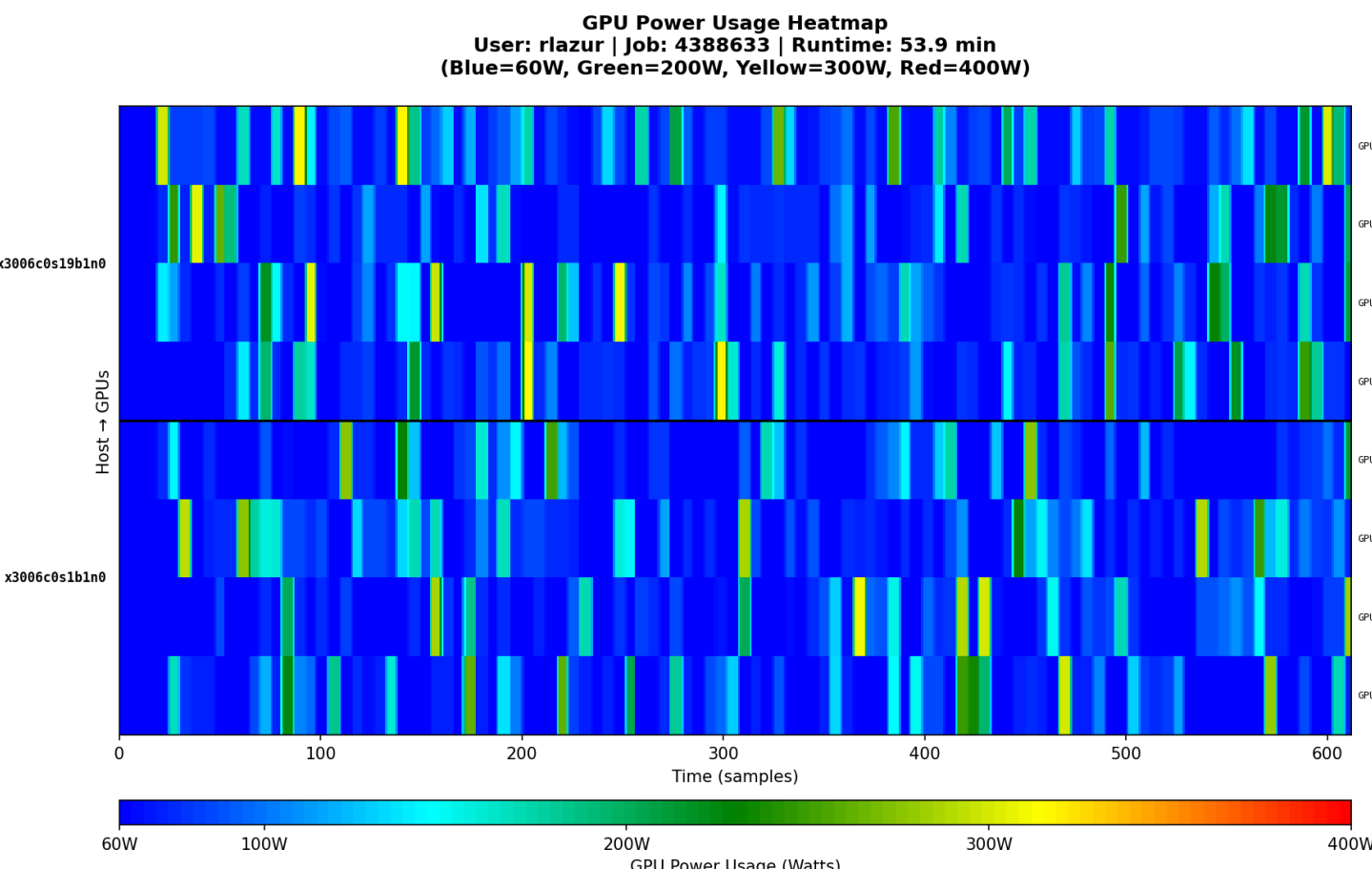
- Transforms sequential data into fractal-like geometric patterns
- Most notably used for genome sequencing



(a) DNA of Human herpesvirus 5 strain TR, complete genome—235681 bp

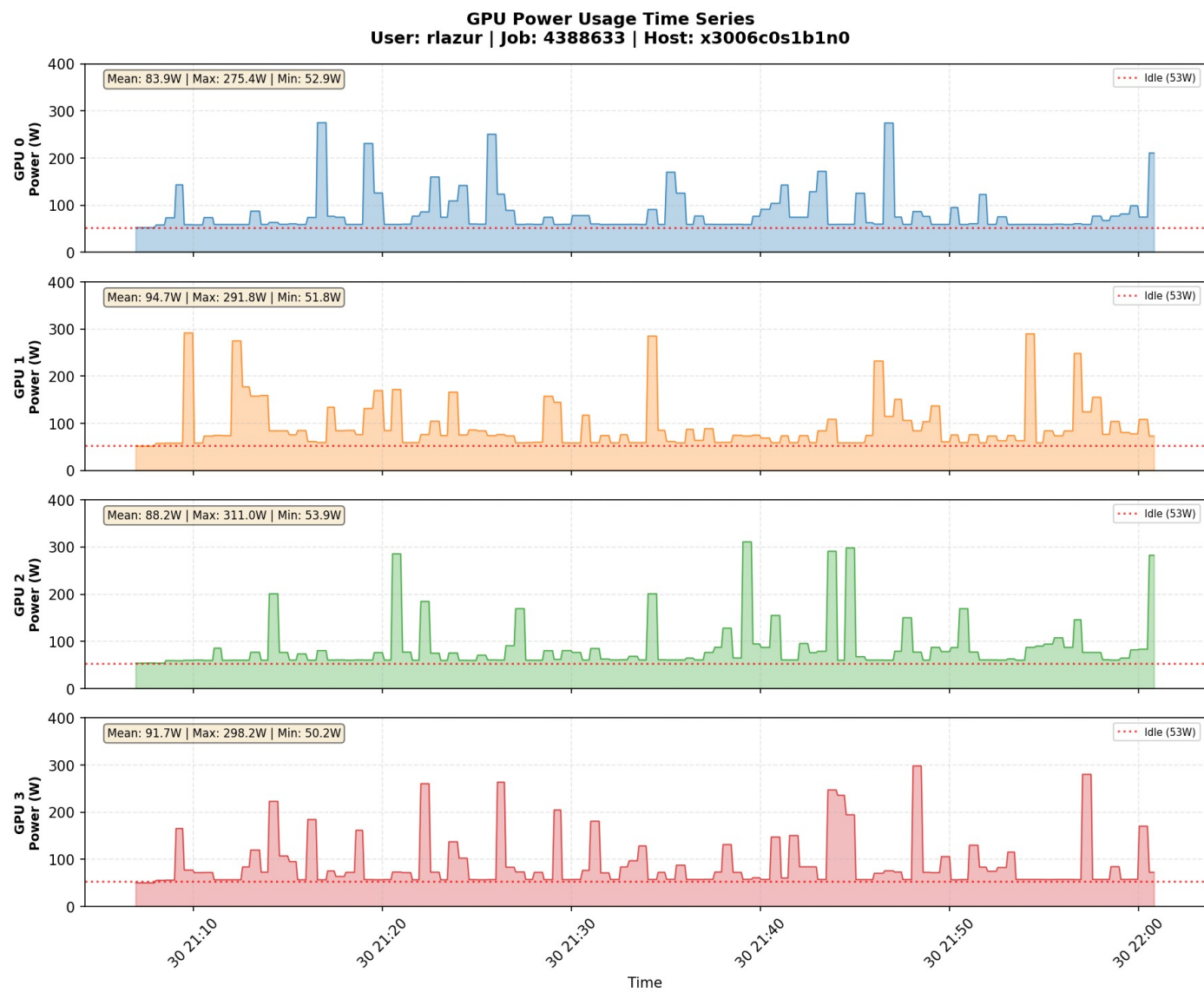


(b) DNA of Chenopodium quinoa cultivar Real Blanca chloroplast, complete sequence, whole genome shotgun sequence—152282 bp



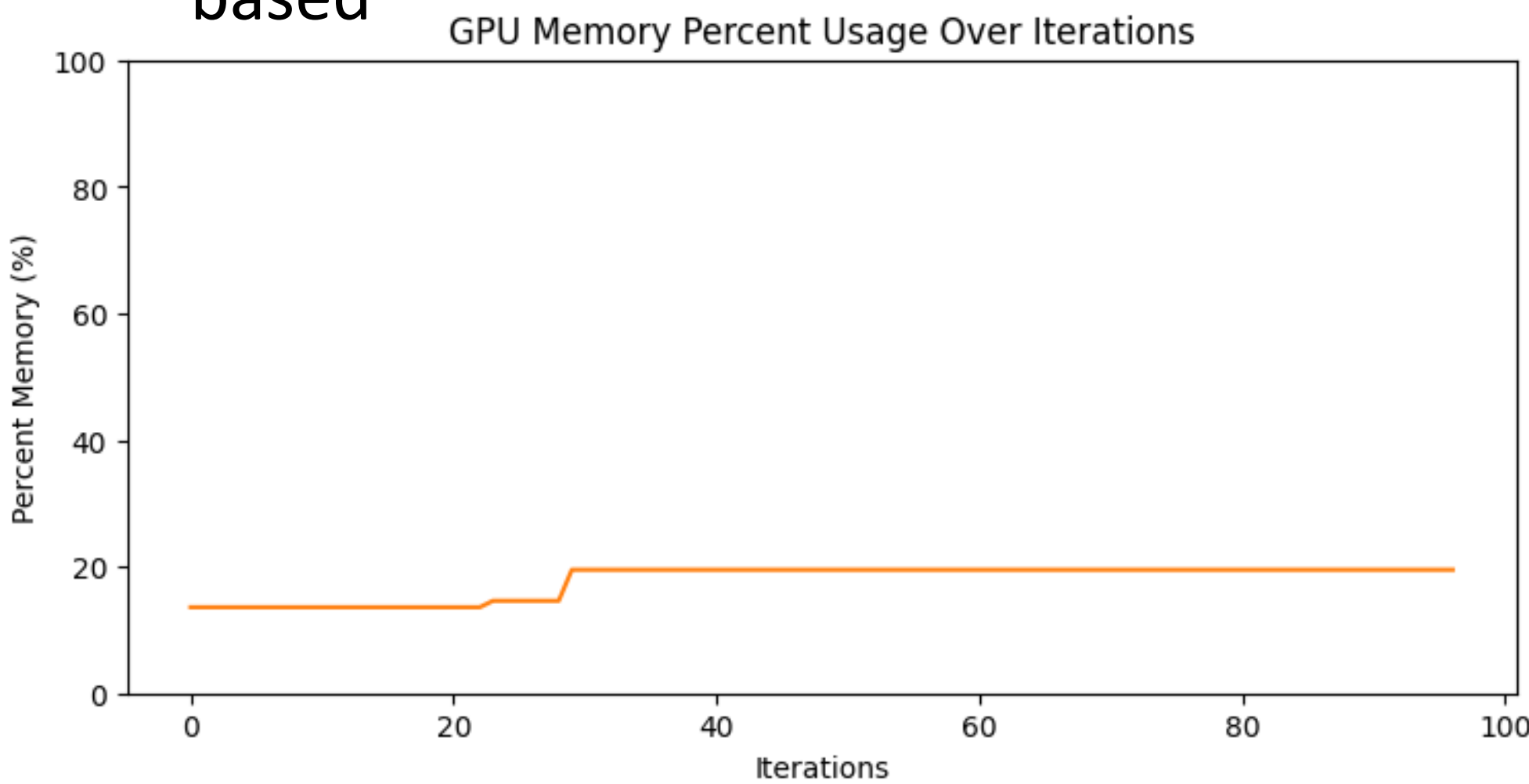
- Steps to adapt resource usage data to CGR:
 - Discretize power usage data (binning)
 - Naïve approach: n evenly spaced bins
 - Should inspect data for better bin ranges

Concerns and Futures Steps



- Future Steps:
 - Investigate other method of time-series analysis
 - Find ways to process data to make it more CGR compatible
 - Decide whether CGR is worth pursuing more

- Number of issues when adapting data to CGR
 - Data scarcity (relative to what is needed)
 - Data uniformity
 - Multiple data streams
 - Binning is unintuitive and implementation-based



References

1. Eunice Y. S. Chan and Robert M. Corless. 2020. Chaos Game Representation. arXiv preprint. ArXiv:2012.09638 [math].
2. H.Joel Jeffrey. 1990. Chaos game representation of gene structure. Nucleic Acids Research, 18(8):2163-2170.
3. Sven Köhler, Lukas Wenzel, Max Plauth, Pawel Böning, Philipp Gampe, Leonard Geier, and Andreas Polze. 2021. Recognizing hpc workloads based on power draw signatures. In 2021 Ninth International Symposium on Computing and Networking Workshops (CANDARW), pages 278–284.